

Article

Optimizing the Redistribution of Surplus Food in the Hospitality Sector: A Paradigm Shift Through the Implementation of Food Donation Systems for a Sustainable Future

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Abstract: Food service is one of the major contributors to food waste globally. The Food Waste Index report has recently showed that the world generates about 1052 million tons of food waste per year, out of which 290 million tons come from food services. This study is focused on food service within the hospitality industry, which plays a significant role in generating food waste in the tourism sector. This study explores an innovative approach to the implementation of food donation systems through hospitality practitioners in tourist destinations for a sustainable future. By employing a blend of qualitative and quantitative research methods, this study analyzes the effectiveness of optimizing the redistribution of surplus food through the implementation of food donation systems through hospitality practitioners and focuses on identifying the existing challenges in the food donation system. This study identifies food waste as a global threat, and the results reveal that implementing various food donation systems can lead to a sustainable future. This study contributes solution strategies for various existing challenges in food donation systems.

Keywords: food waste; food donation; tourism; sustainability; food service



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1. Introduction

Food waste is a significant global challenge, with staggering implications for both the environment and humanity [1]. Minimizing food waste presents a direct solution to increasing food availability for those in need, potentially alleviating starvation and improving the stability of food worldwide [2]. The statistics are sobering, including the World Bank's (2020) estimates that food loss and waste are worth more than 1 trillion USD annually, equivalent to over one-third of the world's total food production [3]. This waste occurs across the supply chain, from production to consumption, utilizing a substantial twenty-eight percent of the agricultural area in the world [4]. According to the Food Waste Index Report 2024 by UNEP, the world generates about 1052 million tons of food waste per year, out of which 290 million tons come from the food service sector [5]. During the year 2022, an alarming twenty-nine percent of the population worldwide faced moderate or severe food insecurity, with approximately 783 million individuals experiencing hunger, increasing from 122 million in 2019 [6].

Food waste significantly contributes to the emissions of greenhouse gases, accounting for an estimated ten percent of total emissions [7–9]. The worldwide greenhouse gas (GHG) emissions reached a new high of 57.4 GtCO₂e in 2022, indicating an increase of 1.2% over the previous year [10]. Fossil CO₂ emissions, which constitute around two-thirds of global GHG emissions, underwent an upswing of 0.8–1.5%. Fluorinated gases recorded a 5.5% growth

rate, while methane and nitrous oxide (N₂O) increased by 1.8% and 0.9%, respectively [8]. For the G20 countries in particular, there was an increase in CO₂ concentration of 1.2 percent, with this change driven by China, India, Indonesia, and the United States, while Brazil, the European Union, and the Russian Federation showed downward trends [4,5]. Together, these countries contribute to about three-quarters (76%) of the worldwide GHG emissions. Consumer waste is the most burdensome at the end of the food items' lifecycle [11]. This is because not only the resources used in the production are wasted, but also because food decomposing in landfills emits methane, a strong global warming gas responsible for climate change [9,10]. The need to address this problem calls for joint efforts among all the concerned parties, including governments, companies, and the people themselves. In order to solve this issue, approaches such as promoting sustainable consumption systems, upgrading food delivery channels, and implementing waste management policies can be employed. By reducing foodstuff losses, we can not only free millions from hunger and improve their nourishment, but also mitigate environmental damage and arrive at an integrated approach to development for everyone [12–24].

The rise in food waste from 2021 to 2022 is alarming, and it emphasizes the need for prompt action to address this problem [4,5]. Campaigns aimed at reducing food waste should be intensified through public awareness, policy actions, and innovative approaches that promote sustainable practices of food utilization and help reduce the environmental and social consequences of food waste. SDG 12.3 focuses on reducing food loss and waste by means of two indicators: the Food Waste Index and the Food Loss Index [5]. The Food Loss Index managed by the FAO measures the losses of major commodities along the supply chains, which lead to but are not included in the retail sector [5]. On the other hand, the Food Waste Index overseen by the United Nations Environment Programme (UNEP) measures the food and inedible parts wasted at the retail and consumer levels, including food services and households [4]. Unlike the Food Loss Index, the Food Waste Index accounts for the total food waste, encompassing both edible and inedible parts [5]. These indicators are crucial for monitoring progress towards the reduction of excessive food loss and waste, fostering more effective sustainable consumption and production patterns and ultimately contributing to the achievement of SDG 12.3 [5].

1.1. Current Status of Food Waste

The Food Waste Index Report (2021) estimated that in 2019, food waste was generated in the enormous amount of 931 million tons across three different sectors, most notably households, the retail sector, and food services, with an average of 121 kg per capita per year [4,5]. The report revealed a huge difference in the 2019 estimate of 931 million tons compared to the 2022 estimate of 1.05 billion tons, which shows a large increase in food waste [5]. This suggests that efforts to reduce food waste may have been somewhat successful, or that there are fluctuations in reporting or estimation methods. Ongoing monitoring and efforts to reduce food waste remain critical for sustainable food systems [5,6].

Food waste and loss, rising food prices, and an increase in the prevalence of food poverty all indicate that the problem is becoming worse [3]. Every step of the food chain—production, mode of transport, storage, processing, marketing, and consumption—experiences losses and waste. High-income nations discard a total of 79 kg worth of food per individual annually, in accordance with the United Nations Food Waste Index Report 2021 [5].

Figure 1 illustrates the global food waste findings from the UN Food Waste Index reports in 2022 across the three sectors analyzed, which show 1.05 billion tons of food waste, amounting to 132 kg per capita per year. Notably, 12 percent of the waste originated from the retail sector, 60 percent from households, and, finally, 28 percent from food services [5].



Figure 1. Estimation of global food waste [5].

Figure 2 reveals the food waste generated by different countries, categorized by region, across the retail, food service, and household sectors. The data highlight that food waste generation is a major global issue. The increase in food waste from 2021 to 2022 is particularly concerning, emphasizing the urgent need for action to address this challenge. Efforts to reduce food waste must be intensified through awareness campaigns, policy interventions, and strategic solutions to promote sustainability in food practices and mitigate the ecological and social implications of food waste [5].

Comparison of Region Wise Food Waste

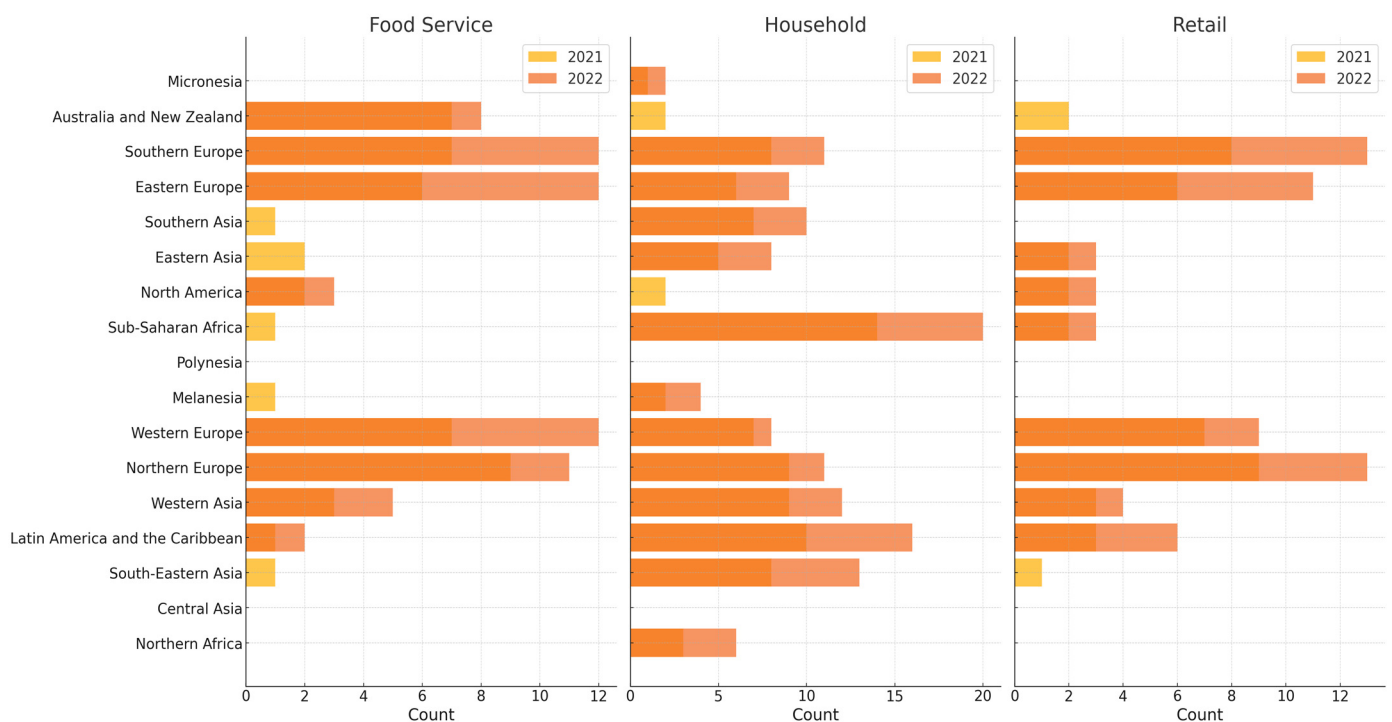


Figure 2. Regional food waste increased from 2021 to 2022 [5].

1.2. Food Waste in India

India is a nation rich in cultural traditions, diverse flavors, and culinary heritage [14]. However, it faces a significant challenge, with around 230 million citizens affected by the issue of food waste [15]. According to the Ministry of Agriculture, food waste in India is estimated to be valued at approximately 500,000,000,000 INR annually [19]. The staggering amount of food wasted daily in a country where millions still go hungry is a sobering and unfortunate reality [25].

In India, the staggering amount of food waste totals 68.7 million tons annually, equating to almost 50 kg per person, as reported by the UNEP's Food Waste Index [4,5]. Food waste is a significant issue that contributes to environmental degradation and the loss of natural resources, primarily through methane emissions from decomposing food in landfills [13]. Similarly, in the United States, 40% of the food produced is squandered, highlighting a global issue of inefficiency in food distribution and consumption [26–30]. Moreover, the 1.34 billion tons of food wasted annually in Asia, with China and India leading, underscores the urgent need for comprehensive solutions to address food waste across the entire supply chain [16]. Findings from the FSSAI in India reveal that one-third of the total food production is lost or wasted before consumption. This alarming issue is attributed to factors such as a lack of awareness and education about the scale of food waste, as well as the absence of strict regulatory policies to prevent food waste [11,12]. Sustainable consumption practices and well-monitored food waste reduction strategies must be implemented to mitigate this crisis [18,19]. Currently, food waste poses an international threat, especially in India, affecting environmental, economic, and societal aspects. According to the Food and Agriculture Organization (FAO), approximately 1.3 billion tons of food produced globally for consumption is wasted annually. The consequences of such massive food waste include greenhouse gas emissions, depletion of natural resources, food insecurity, and exacerbation of poverty [17,18].

According to the Global Hunger Index 2021, India ranked 101st out of 116 nations, and in 2023, its position dropped to 111th out of 125 nations [19–21]. This is a concerning trend, especially given the high levels of food waste in the country, even in such a deplorable situation [20,21]. Food services and retail establishments are significant contributors to this issue, as highlighted in the United Nations Environment Programme Food Waste Index Report 2021 [6,9].

One effective solution to reduce food waste is food donation, which plays a crucial role in sustainable development by reducing food waste, alleviating hunger, and fostering social inclusion [26,27]. Food donations redistribute surplus food to those in need, addressing both environmental and social challenges. Supporting such initiatives can help communities adopt sustainable practices and enhance societal well-being [6].

Figure 3 illustrates the percentage distribution of food waste in India: 61% of food waste originates from households, significantly surpassing the contributions from food services (26%) and retail outlets (13%). These data underscore the urgent need for targeted strategies to minimize food waste. Addressing this issue can lead to significant improvements in waste management and sustainability efforts, aligning with India's broader sustainability goals [4,6,8].

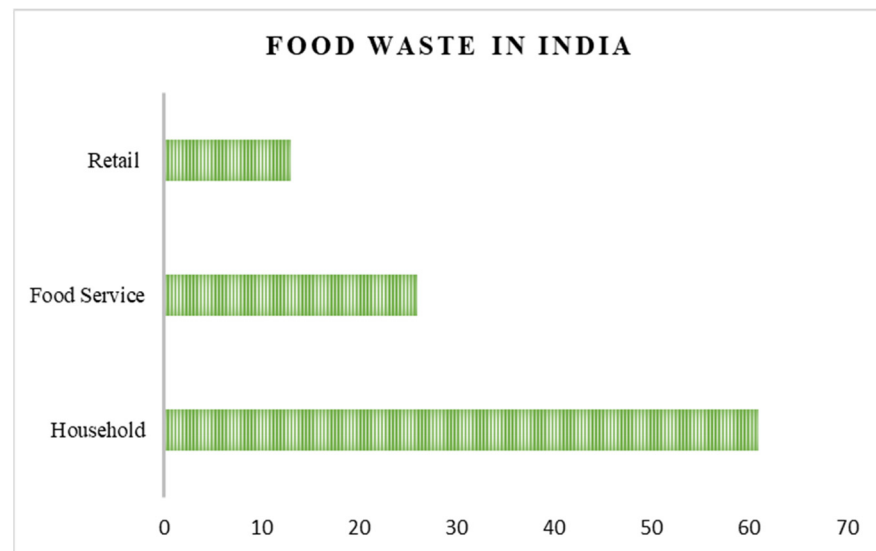


Figure 3. UNEP food waste index 2021 [5].

2. Literature Review

2.1. Sustainability and Waste of Food

Food loss and food waste (FLW) are terms often used interchangeably in the literature, referring to the loss or waste of food at various stages of the food supply chain, from production to consumption [22]. Some researchers define food loss as the loss of food during preparation and handling in post-harvest management, while food waste refers to the total spoilage of food resulting from consumption patterns and inefficiencies in distribution channels [24]. Additionally, food waste is often considered a subset of food loss, specifically involving food that is intentionally discarded, neglected until spoiled, or deemed unfit and unsuitable for human consumption [25]. Despite increasing attention to food spoilage in recent years, there remains a need for a more comprehensive understanding of the issue, hindered by limitations in time, geographical coverage, and data consistency [27]. Research indicates that banqueting and food service provisions contribute significantly to FLW, potentially impeding the achievement of the United Nations Sustainable Development Goals (SDGs). A recent systematic literature review employing the social practice theory (SPT) highlights the importance of exploring practices to reduce organic waste in kitchens and improve administrative efficiency. It emphasizes the need for a hands-on methodological approach, which could facilitate future learning and inform policy development. Addressing these challenges effectively requires urgent and targeted research efforts [28–30].

2.2. Management of Food Waste and the Waste Hierarchy Model

Anaerobic digestion, animal feed, recycling, energy recovery, and landfill disposal are common methods of managing food waste; however, the prevention of food waste is often overlooked [29]. According to the hierarchy principles published by the European Commission, prevention, recycling, reuse, recovery, and disposal are ranked as the most to least desirable solutions for addressing food waste [25]. This model has been widely adopted and modified by organizations, scholars, researchers, and practitioners into a conceptual framework for sustainable waste management and reduction [26].

The US Environmental Protection Agency (EPA) has developed a model approach for recycling discarded food within the waste hierarchy framework, providing guidance on food waste prevention and management strategies for individuals and organizations [30–32]. This model emphasizes four essential principles of food waste management, employing

cost-effective strategies to reduce environmental impact and promote sustainability [27]. The four key principles of this approach are as follows. First, prevention and reduction, which focus on minimizing waste generation through responsible consumer behavior. Second, reuse, which involves finding alternative uses for edible food products that would otherwise be discarded, thereby maximizing food utilization. Third, recycling, which reduces waste by converting it into animal feed or fertilizer, supporting sustainable agricultural practices. Fourth, recovery, which uses techniques such as anaerobic digestion to convert food waste into bioenergy, thereby reducing pollution. The final options for disposing of waste—landfilling or incineration—are considered last-resort measures. By following this hierarchy, we can significantly reduce food loss, conserve resources, and minimize environmental harm, thereby adopting more sustainable methods for consuming and managing food waste [27–35].

Despite its benefits, the food waste hierarchy model has limited applications in the hospitality sector for food waste management. The use of this model can significantly minimize waste during various stages of food distribution and hotel operations. The authors asserted that food waste hierarchy practices can greatly assist hotels in managing this issue effectively [13]. These practices include preventing food waste from occurring, reusing food waste, and maximizing the utilization of all available food resources. Such efforts contribute to eliminating waste, fostering a more sustainable and efficient system for food management in hotels [27,28].

The evaluation of food waste within restaurants in developed nations requires further enhancement [36–41]. A significant gap exists between sustainable ideas and recycling activities in the hospitality industry, particularly in developing and transitioning countries [42]. While the majority of food waste research focuses on restaurants, there are substantial research gaps across various types of dining establishments, resulting in diverse food waste management techniques [43–46]. The food waste generated by the hotel industry is comparatively lower; however, this aspect requires further investigation and the implementation of necessary actions. Food waste in restaurants remains a critical issue that demands deeper exploration and effective solutions [47–51]. Bridging these gaps and overcoming obstacles is crucial for implementing successful food waste management strategies in the hotel industry, thereby fostering a sustainable future [46].

2.3. Food Waste Management in the Hospitality Industry

Effective food waste management strategies and sustainability measures implemented by conscientious hospitality and food service organizations were highlighted in prior studies. During the planning stage, hotels can proactively reduce food waste by rethinking kitchen processes [52]. Waste can be minimized during the handling stage by conducting routine storage inspections to ensure adherence to the first-in-first-out (FIFO) inventory system [53]. Additionally, unsold, underused, or excess food can be redistributed by offering it for sale at a discount, donating it to employees or nonprofits, or utilizing it in employee cafeterias [54]. Food waste in restaurants can also be reduced by switching from buffet-style to à la carte service [48]. Inedible food scraps can be composted, converted into fish meal, or repurposed as pet food through doggy bags for customers' pets. Measuring food waste is crucial for effectively reducing it [55–59]. However, the management of food waste in hotels remains fragmented, despite the increasing body of research on food waste in the hospitality sector and food provision operations in recent years [60,61].

Although good practices exist, establishments in the hospitality sector and food service industries, particularly hotels and restaurants, continue to face significant challenges [50,62]. For instance, insufficient resources to manage food waste, evolving restaurant recycling techniques, and the influence of corporate policies and regulations pose obstacles to effective

tive food waste prevention measures. Notably, studies by Liumpetch and Srisuwannaket, as well as by Srijuntrapun et al., highlight these issues and their implications [53].

3. Objectives of the Study

1. To analyze the effectiveness of implementing food donation systems within the hospitality sector.
2. To examine the existing challenges in food donation systems within food services in the hospitality sector.

4. Materials and Methods

4.1. Study Location

The study adheres to the STROBE guidelines for reporting observational studies, ensuring clarity and transparency in its methodology and findings. The research was conducted in the Alappuzha district of Kerala, India, a prominent tourism destination. This location was selected based on visitor arrivals, as reported in the Indian Tourism Statistics 2023 [63].

4.2. Research Design

The study employed both qualitative and quantitative research approaches. Qualitative analysis was conducted using content analysis. The population for the study comprised food service establishments in the hospitality sector, while the sample units were hospitality practitioners in the tourism destination of the Alappuzha district. The sample size, determined using the Taro Yamane method, was 300 respondents. The variables for the questionnaire were adapted from a literature review and content analysis conducted using NVivo and Taguette. A structured questionnaire employing a seven-point Likert scale was used to quantify the data. The research design is both descriptive and analytical in nature. Primary data were collected through a survey method based on cross-sectional analysis, using the computer-assisted personal interviewing (CAPI) technique, which involved face-to-face interviews with respondents. The study adopted a proportionate stratified random sampling technique for data collection.

4.3. Statistical Method

Percentage analysis, Levene's test, and multiple regression were employed as statistical methods to test the hypotheses. The reliability of the variables measuring the effectiveness and challenges of the food donation system was found to be between 0.900 to 0.904, indicating high and strong internal consistency, making it suitable for further analysis. Normality was assessed using a P-P plot and a histogram of the regression analysis, confirming a normal distribution. The study utilized SPSS version 26 as the statistical tool for data analysis.

4.4. Measurement of Variables

The questionnaire is divided into three sections include demographic profile (Appendix A), factors contributing to the effectiveness of food donation system (Appendix B) and the implementation of the food donation system (Appendix C). This study measured the variables of reducing food waste, promoting sustainability, and collaboration with food banks using established methods from relevant literature. The implementation of the food donation system was assessed based on its ability to reduce food waste, promote sustainability, and achieve partnership goals. Sustainability was evaluated through a Likert scale, following approaches similar to those used by Schneider and Thyberg and Tonjes, to measure environmental and social benefits. Partnership goals were assessed using interviews and questionnaires, as described by Garrone et al., focusing on the frequency, quality, and effectiveness of partnerships. Statistical tests were employed to

ensure reliability and validity. Challenges related to food donations were measured using variables such as logistics, awareness, technological infrastructure, quality and safety, and collaboration. Logistics challenges assessed the efficiency of transportation, storage, and distribution management, while awareness challenges evaluated public and donor understanding of food donation benefits. Technological infrastructure challenges examined the integration of advanced technologies such as IoT and blockchain for improved management. Collaboration challenges focused on the efficiency of stakeholder coordination, and quality and safety challenges ensured that food safety and quality were maintained during collection and distribution [64–74].

5. Analysis and Results

5.1. Demographic Profile of the Respondents

Table 1 describes the demographic profile of the respondents. The majority of the sample (77%) was aged 40–50, indicating a middle-aged adult category. The gender category shows slightly more males (72%) than females (28%) in the sample. The highest percentages of respondents were from cafes (26%) and restaurants (22%), indicating a high potential for food donation initiatives in these environments due to their large size and frequent surplus. Combined, casual hotels (18%) and star hotels (17%) represented 35% of the sample. These establishments might have structured processes for food preparation and disposal, making them suitable for systematic food donation programs. Juice bars (5%), bakeries (4%), coffee shops (6%) represent a smaller proportion. Catering services (8%) often deal with large quantities of food for events, presenting a unique opportunity for planned food donation efforts, especially post-event. Managers (59%) form the largest group of respondents, suggesting that managerial perspectives are predominantly represented. Staff (20%) and chefs (18%) also have substantial representation, providing insights from operational roles. Owners (3%) are the least represented group, indicating fewer insights from ownership perspectives.

Table 1. Demographic profile of the respondents.

	Categories	Frequency	Percent	Cumulative Percent
Age	18 years to 28 years	18	6	6
	29 years to 39 years	42	14	20
	40 years to 50 years	177	57	77
	51 years to 61 years	48	16	93
	≥62 years	21	7	100
Gender	Total	300	100	
	Male	216	72	72
	Female	84	28	100
	Other	0	0	
Occupation	Total	300	100	
	Resorts	39	13	13
	Juice bars	15	5	18
	Bakeries	12	4	22
	Coffee shops	18	6	28
	Catering services	24	8	36
	Star hotels	51	17	53
	Casual hotels	54	18	71
	Cafes	21	7	78
	Restaurants	66	22	100
Repondents	Total	300	100	
	Owners	9	3	3
	Staff	60	20	23
	Chefs	54	18	41
	Managers	177	59	100
	Total	300	100	

5.2. Hypothesis

H0. *There is no significant relationship between factors contributing to the effectiveness of food donation system and implementing the food donation system in the hospitality sector.*

The results from Levene's test for homogeneity of variance in Table 2 indicate that the assumption of equal variances was met ($p = 0.683$). This suggests that the variances are equal across the groups, and therefore, the data are suitable for further parametric analysis.

Table 2. Test of homogeneity of variance.

Levene's Test	df1	df2	Sig.
0.465	4	1508	0.683

In the model summary described in Table 3, the multiple regression analysis showed a strong positive correlation ($R = 0.742$) between the predictors and the effectiveness of the food donation system. The R squared value of 0.550 indicated that approximately 55% of the variance in the effectiveness of the food donation system can be explained by the independent variables, which are food waste reduction, promotion of sustainability, and partnership collaboration. The adjusted R squared value of 0.546 was slightly lower but still indicated a good fit of the model. The standard error of the estimate (1.289) reflects the average distance from the observed values to the regression line. Additionally, the Durbin–Watson statistic (1.846) indicated no significant autocorrelation in the residuals, suggesting the independence of errors. All these variance indicators show a good model fit.

Table 3. Model summary.

Model	R	R Squared	Adjusted R Squared	Std. Error of the Estimate	Durbin–Watson
1	0.742	0.550	0.546	1.289	1.846

Table 4 illustrates the ANOVA results of the regression analysis, providing evidence of the overall significance of the regression model. The F-statistic (120.723) was significant ($p = 0.000$), indicating that the regression model provided a good fit to the data and that the predictors together reliably predict the effectiveness of the food donation system. Thus, the null hypothesis stating that “There is no significant relationship between factors contributing to the effectiveness of food donation system and implementing the food donation system in the hospitality sector” was rejected.

Table 4. ANOVA.

Model 1	Sum of Squares	df	Mean Square	F	Sig.
Regression	601.618	3	200.539	120.723	0.000
Residual	491.702	296	1.661		
Total	1093.320	299			

Table 5 illustrates the regression coefficients, providing details on each predictor's contribution. The unstandardized coefficients indicate the effect of each independent variable on the dependent variable. The constant term ($B = 0.827$, $p = 0.000$) was the expected mean value of the dependent variable when all the predictors were zero. Food waste reduction ($B = 0.164$, $p = 0.005$), promotion of sustainability ($B = 0.204$, $p = 0.000$), and partnership collaboration ($B = 0.443$, $p = 0.000$) all significantly positively predicted the effectiveness of the food donation system. The standardized coefficients (Beta) showed

that partnership collaboration had the strongest influence (Beta = 0.451), followed by food waste reduction (Beta = 0.269) and promotion of sustainability (Beta = 0.217).

Table 5. Coefficients.

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistic	
	B	Std. Error	Beta			Tolerance	VIF
(Constant)	0.827	0.217		3.810	0.000		
Food waste reduction	0.164	0.057	0.269	2.854	0.000	0.431	2.319
Promotion of sustainability	0.204	0.056	0.217	3.657	0.000	0.431	2.321
Partnership collaboration	0.443	0.052	0.451	8.516	0.000	0.543	1.842

The collinearity statistics (tolerance and VIF) indicated no multicollinearity issues among the predictors, as the tolerance values were above 0.1 and the VIF values were below 10. Specifically, the tolerance values for food waste reduction and promotion of sustainability were 0.431, with the VIF values of 2.319 and 2.321, respectively, while partnership collaboration had a tolerance of 0.543 and a VIF of 1.842. Hence, the null hypothesis stating that “There is no significant relationship between factors contributing to the effectiveness of food donation system and implementing the food donation system in the hospitality sector” was rejected.

Implementing a food donation system in the hospitality sector significantly enhances its effectiveness by reducing food waste and promoting sustainability. Hotels and restaurants can repurpose surplus food, which otherwise goes to waste, to support community needs, thereby fostering social responsibility. This system not only aids in feeding the hungry, but also improves the sector’s public image and compliance with waste management regulations. Additionally, it can lead to cost savings in waste disposal and enhance employee engagement through involvement in charitable activities, ultimately contributing to a more sustainable and socially responsible hospitality industry.

Results of the regression analysis reveal that effective food donation enhances efforts, emphasizing the crucial role of food donations in sustainable development. Food donation systems can help reduce hunger, promote responsible consumption, and contribute to several SDGs, such as Zero Hunger (SDG 2), Responsible Consumption and Production (SDG 12), and Partnerships for the Goals (SDG 17). The reduction of food waste through donations is a significant aspect. By redirecting surplus food to those in need, both food insecurity and environmental issues related to food waste are addressed. This approach not only conserves resources, but also reduces the carbon footprint associated with wasted food. Effectiveness of the food donation system often depends on strong partnerships between various stakeholders, including non-profits, businesses, government agencies, and community organizations. Collaborative efforts ensure a more efficient and widespread distribution network, enhancing the reach and impact of food donation programs.

Figure 4 presents the standardized residuals plotted against the dependent variable, which measures the effectiveness of the food donation system, and the independent variable factors contributing to food donation implementation, including reduction of food waste, promotion of sustainability, and achievement of partnership goals. The normal P–P plot of regression standardized residuals shows that the residuals closely follow the diagonal line, indicating they are approximately normally distributed. This suggests that the error terms of the regression model for the effective food donation system and factors contributing to food donation adhere to a normal distribution. The slight deviations at the tails are

minor and do not significantly impact the overall normality assumption. Thus, the model's assumption of normally distributed residuals is reasonably satisfied.

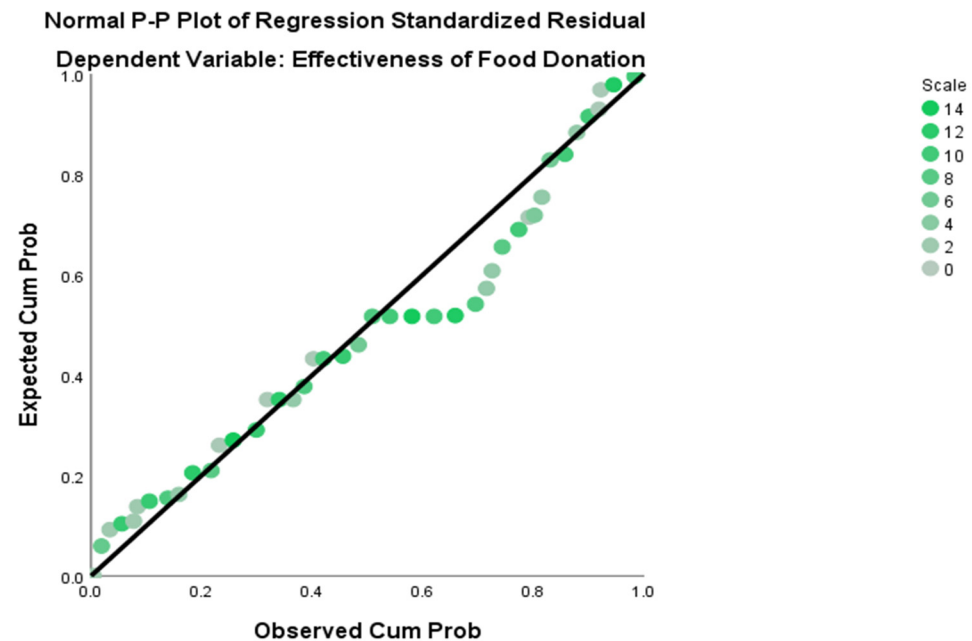


Figure 4. Normal P–P plot of the regression of food donation effectiveness.

Figure 5 displays a histogram overlaid with a normal curve, indicating an approximately normal distribution.

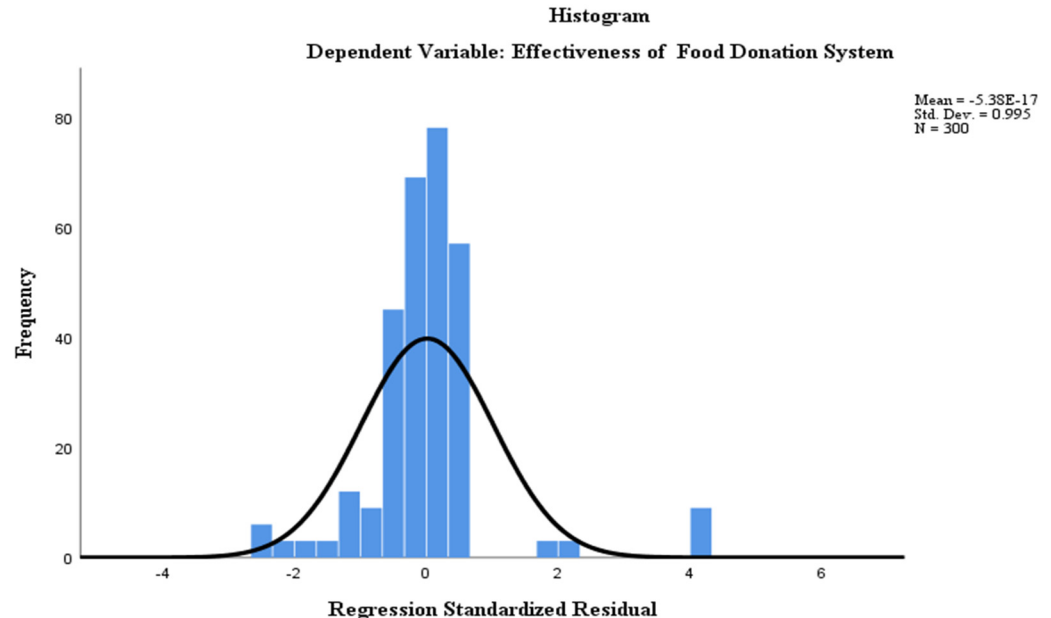


Figure 5. Histogram of the normal distribution of the food donation system's effectiveness.

5.3. Hypothesis 2

H1: *The existing challenges do not affect the implementation of food donation systems in the hospitality sector.*

Levene's test for homogeneity of variance, as presented in Table 6, yielded a significance value of 0.811, indicating that the variances were equal across the groups. This result

confirms that the assumption of homogeneity of variance was met, validating the use of further parametric tests in the analysis.

Table 6. Test of homogeneity of variance.

Levene Statistic	df1	df2	Sig.
0.532	5	1465	0.811

The model summary in Table 7 indicates a strong positive correlation ($R = 0.726$) between the independent variables (logistic challenges, technology infrastructure, awareness, collaboration, and quality and safety) and the dependent variable (effectiveness of implementing a food donation system). The R squared value of 0.528 reveals that approximately 52.8% of the variance in the effectiveness of the food donation system could be explained by these independent variables. The adjusted R squared value of 0.520, which adjusts for the number of predictors, still indicated a good fit of the model. The standard error of the estimate was 1.325, reflecting the average distance that the observed values fell from the regression line. Additionally, the Durbin–Watson statistic of 1.970 suggested that there was no significant autocorrelation in the residuals, supporting the assumption of independence of errors. All these variances indicate a good model fit.

Table 7. Model summary.

Model	R	R Squared	Adjusted R Squared	Std. Error of the Estimate	Durbin–Watson
2	0.726	0.528	0.520	1.325	1.970

Table 8 illustrates the ANOVA results of the regression analysis. The ANOVA table supports the overall significance of the regression model, with an F-statistic of 65.667 and a significance value (p -value) of 0.000, which is less than the significance level at 1 percent recorded in the output. This highly significant F-test result indicates that the regression model is a good fit for the data and that the predictors together reliably predict the effectiveness of the food donation system, leading to the rejection of the null hypothesis that “The existing challenges do not affect the implementation of the food donation system in the hospitality sector.”

Table 8. ANOVA.

Model 1	Sum of Squares	df	Mean Square	F	Sig.
Regression	576.818	5	115.364	65.667	0.000
Residual	516.502	294	1.757		
Total	1093.320	299			

Table 9 illustrates the regression coefficients and provides detailed insights into the contribution of each predictor. The constant term ($B = 1.031$, $p = 0.000$) represents the expected mean value of the dependent variable when all predictors are zero, that is, the effectiveness of the food donation system. Logistic challenges ($B = 0.229$, $p = 0.001$) are shown to be a significant positive predictor of the effectiveness of the food donation system. Similarly, technology infrastructure ($B = 0.178$, $p = 0.005$), awareness ($B = 0.173$, $p = 0.000$), collaboration ($B = 0.378$, $p = 0.000$), and quality and safety ($B = 0.380$, $p = 0.000$) all significantly and positively influenced the effectiveness of the food donation system. Among these, collaboration (Beta = 0.398) and quality and safety (Beta = 0.405) had the strongest influence, as indicated by their standardized coefficients (Beta).

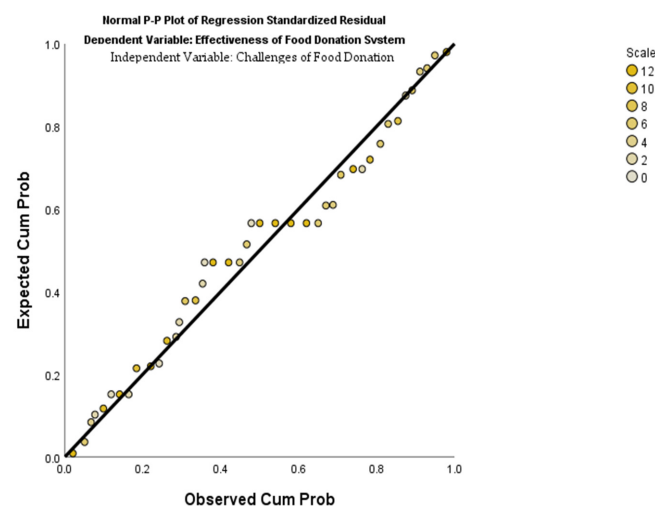
Table 9. Coefficients.

Model	Unstandardized Coefficients		Standardized Coefficients	T	Sig.	Collinearity Statistic	
	B	Std. Error	Beta			Tolerance	VIF
(Constant)	1.031	0.256		4.031	0.000		
Logistic	0.229	0.068	0.227	3.337	0.001	0.346	2.888
Technology infrastructure	0.178	0.102	0.186	1.748	0.005	0.142	7.044
Awareness	0.173	0.198	0.273	0.746	0.000	0.169	5.923
Collaboration	0.378	0.083	0.398	4.561	0.000	0.211	4.729
Quality and safety	0.380	0.066	0.405	5.771	0.000	0.327	3.061

The collinearity statistics reveal that while most predictors had acceptable tolerance values (above 0.1) and VIF values (below 10), there was some indication of multicollinearity, especially with technology infrastructure (VIF = 7.044) and awareness (VIF = 5.923). These high VIF values suggest that there may be redundancy among the predictors, and further investigation into multicollinearity is warranted.

The regression analysis provided strong evidence that logistic challenges, technology infrastructure, awareness, collaboration, and quality and safety significantly impact the implementation of the food donation system in the hospitality sector. The model explains 52.8% of the variance in effectiveness, and all predictors are significant. Therefore, the null hypothesis was rejected, affirming that the existing challenges affect the effectiveness of implementing the food donation system. This analysis highlights the critical areas that need to be addressed to improve the implementation of food donation systems in the hospitality sector. A necessary strategic solution must be implemented for an effective food donation system.

Figure 6 shows a normal P–P plot of regression standardized residuals against the dependent variable, effectiveness of food donation, and independent variables. The normal P–P plot of regression standardized residuals evaluated the normality of residuals in a regression model, assessing the effectiveness of a food donation system against its challenges. Points closely following the diagonal line indicate normally distributed residuals, confirming the model’s validity. This suggests that the regression model fit well, supporting the conclusion that challenges such as logistics, awareness, collaboration, and technological infrastructure significantly impact the food donation system. The alignment of data points with the diagonal line ensures that the model’s assumptions were met, making the findings on improving food donation practices reliable.

**Figure 6.** Normal P–P plot of regression against the challenges of food donation.

6. Strategic Solution for Effective Food Donation Systems

Traditional food donation practices face several challenges, including logistics, awareness, collaboration with stakeholders, technological infrastructure, and ensuring the quality and safety of donated food. To overcome these obstacles, the article provides strategic solutions. Effective management of surplus food through donation holds significant potential to address food poverty while simultaneously reducing food waste. The article highlights various strategies and innovative methods that can enhance the efficiency and effectiveness of food donation systems, contributing to a more sustainable future. These solution strategies are not only tailored to address the identified challenges, but can also be adapted and tested to overcome other related issues in food waste management systems. The following are the proposed solution strategies for the effective implementation of food donation initiatives.

6.1. AI Tracking System for Food Donations

An AI-powered tracking system can identify surplus food from various sources, such as restaurants, hotels, events, and households, using advanced intelligence applications [75]. By analyzing patterns of surplus food generation, AI can predict when and where surplus food is likely to occur and suggest optimal routes for redistribution to minimize waste [76]. This technology can also connect surplus food with organizations or individuals in need, making the donation process more efficient and significantly reducing food waste. A practical example of an AI surplus food tracking system would be a mobile application that allows hoteliers to share information about their excess food items, including quantity and type. The app could employ AI algorithms to analyze historical data and current trends, predicting when hotel supplies are likely to become surplus. When surplus food is identified, the system can automatically notify nearby registered charity groups or food banks within its network. Additionally, AI can recommend the most efficient ways to relocate the surplus, considering real-time traffic conditions and logistical factors. This ensures that excess food is quickly and conveniently delivered to where it is most needed, reducing waste and supporting those in need [77].

SecondBite, a non-profit organization based in Australia, has developed a Food Rescue Robot to address both food waste and hunger. This AI-driven robot serves a dual purpose as a food collector and distributor [75]. It identifies surplus food from donors in the food service sector, including restaurants, supermarkets, hotels, and caterers, and efficiently delivers it to community organizations and local charities [65]. The Food Rescue Robot optimizes the food donation process, from the collection of excess food to determining the most efficient distribution routes, while ensuring safety and timely delivery. By leveraging AI, the robot makes real-time decisions to maximize the effectiveness of food rescue operations. This technology not only minimizes food waste, but also ensures that surplus food is utilized to nourish those in need, benefiting both the environment and society. The innovative approach of SecondBite's Food Rescue Robot exemplifies how AI can positively impact sustainability and social welfare [75,76].

A non-profit organization called Food Rescue opened in the United States to operate a food recovery network to connect surplus food from various establishments with local organizations that provide meals to people in need, utilizing technological advancements. The organization employs mobile applications and online platforms as tracking systems to identify surplus food and signal its availability, facilitating the food donation process. Food Rescue aims to curb food waste by enabling food service providers to upload information about excess food through the application. This information is then made available to local non-profit organizations, allowing them to collect and distribute the surplus to those in need. Additionally, the application allows donors to track their contributions, offering insights

into how their donations have positively impacted others. By leveraging technology, Food Rescue ensures an efficient and transparent system to reduce food waste and address hunger [75–77].

6.2. Smart Food Donation Refrigerator

Food waste is a major global threat, and strategic initiatives such as the introduction of smart food donation refrigerators aim to tackle this issue effectively. These refrigerators are strategically placed in front of supermarkets, malls, and other public locations, enabling citizens and establishments to donate surplus food. The donated food can then be accessed by individuals in need, free of charge. This system not only promotes food security, but also minimizes food waste, fostering sustainable food practices. Community participation is essential for the success of this initiative, as donations of excess food to these refrigerators ensure proper food waste management. This innovative approach is convenient and allows donors to contribute surplus food effortlessly, without logistical challenges. Dubai has successfully implemented this system with the introduction of smart food donation freezers, leveraging technological advancements to address food waste effectively. The initiative, which operates daily, has been widely appreciated and serves as a model for other nations to adopt. Expanding this system globally could significantly reduce food waste and promote sustainability [78,79].

6.3. Different Applications of AI in Food Waste Management

The various applications of AI in food waste management significantly contribute to the minimization of food waste. These AI-driven solutions are illustrated below as effective strategies for addressing food waste.

Food surplus prediction: AI can analyze data such as weather, sales trends, and historical patterns to predict food surplus. This enables businesses and organizations to manage their inventory more efficiently, reducing waste [55,59].

Smart inventory management: AI-powered systems monitor inventory levels in real time, automatically adjusting orders and production schedules to minimize waste [56–60].

Food quality assessment: AI has the capability to assess food quality, predicting which items are likely to spoil faster. This allows businesses to use or distribute these products before they become waste [55–60].

Optimized distribution: AI algorithms optimize food distribution by considering factors such as expiry dates, delivery routes, and demand predictions. This reduces waste across the supply chain [59,60].

Food donation matching: AI can efficiently match surplus food from businesses with food banks or charities, ensuring excess food reaches those in need instead of being wasted [55–60].

Waste tracking and analytics: AI can analyze food waste data to identify trends, patterns, and areas requiring improvement, providing actionable insights for waste reduction [55–60].

6.4. Involvement and Participation of Youth Volunteers from Educational Institutions in the Food Donation System

Students, as the next generation, play a vital role in shaping the future of our society and environment. Implementing a food donation system through youth volunteers offers a significant opportunity to instill a sense of social responsibility and empathy among students. By actively participating in food donation activities, students can develop in-depth knowledge and awareness of issues such as hunger and food waste, empowering them to become agents of change in their communities. The involvement of students in food donation initiatives enhances their personal development and fosters a mindset of community service, which is a highly positive attribute. Through community service,

students have the chance to cultivate leadership skills, empathy, and a sense of purpose. The food donation system driven by youth volunteers is a transformative initiative that enables students to contribute toward safeguarding the nation's sustainability goals. These experiences can leave a profound and lasting impact on their lives and future careers, shaping them into responsible and compassionate individuals [80].

Through their involvement and active participation in food donation programs, students can make substantial contributions to achieving the Sustainable Development Goals (SDGs) while fostering a culture of sustainability and social responsibility. In India, the National Service Scheme (NSS), a flagship program under the Ministry of Youth Affairs and Sports, aims to develop the personality and character of students through community service. Adopting a strategic plan to integrate a food donation system into such initiatives can further enhance its impact and align with broader sustainability objectives [81–83].

Volunteering aims to “provide community service and foster community development by educating society while promoting personal growth among students in educational institutions.” Youth volunteers from educational institutions should conduct awareness campaigns to educate communities about food waste and sustainable practices. These campaigns help increase awareness of the environmental impact of food waste and encourage responsible behavior. Sustainability is a critical component of the donation process. Awareness campaigns can help establish and maintain long-term partnerships with donors and stakeholders, ensuring the sustainability of the food donation system and effectively addressing food waste. The implementation of a food donation system through volunteers can yield significant positive outcomes, including reduced food waste, improved nutrition, enhanced community engagement, and heightened social responsibility. Ultimately, this initiative contributes to reducing food waste and advancing the Sustainable Development Goals (SDGs) [81–87].

The plan for implementing a food donation system in the hospitality sector involves collaboration between donors and youth volunteers. Capable donors, such as those in the hospitality industry, contribute surplus food with the aim of reducing food waste, highlighting the importance of initiating a structured food donation system. Youth volunteers play a pivotal role in collecting surplus food from donors, emphasizing their active involvement in the process. In storage and handling, once the food is collected, it is crucial to ensure proper storage and handling to maintain its quality and safety. Youth volunteers oversee proper food storage practices within the volunteer unit warehouse. Coordinators play a significant role in organizing the collection and distribution of food, ensuring seamless coordination among volunteers and local NGOs. The collected food is distributed to local communities in need through NGOs and community centers, emphasizing the direct and positive impact of the food donation system on beneficiaries. These beneficiaries, often underprivileged communities, receive nutritious food, underscoring the transformative impact of the initiative on their lives. The goal of volunteers in the food donation system extends beyond simply delivering food. It includes ensuring community service, raising awareness, and fostering long-term impact. Regular monitoring and evaluation of the process are essential to ensure effectiveness and identify areas for improvement, highlighting the importance of continuous refinement in the food donation system. Moreover, volunteers are acutely aware of the significant amount of food wasted in the nation, motivating them to actively contribute to mitigating this issue while promoting a culture of sustainability and responsibility [81–87].

Table 10 presents a list of youth volunteer organizations from various countries worldwide. These organizations offer valuable opportunities for young people to participate in community service, cultivate leadership skills, and contribute positively to society.

Table 10. Youth volunteer organizations.

Sl. no.	Service Scheme	Purpose of the Organization
1.	National Service Scheme (NSS)—India	A program under the Indian government that engages students in community service and nation-building activities [80].
2.	Peace Corps—United States	A volunteer program run by the US government that sends American volunteers abroad to work on development projects [82].
3.	Community Service Volunteers (CSV)—United Kingdom	A volunteering and training organization in the UK that provides opportunities for people to volunteer in various community projects [83].
4.	Jugendfreiwilligendienste (Youth Volunteer Services)—Germany	Offers young people in Germany the opportunity to engage in voluntary service in various social, cultural, and environmental projects [86].
5.	Youth Challenge International (YCI)—Canada	Offers volunteer programs for young people to work on sustainable development projects in various countries [84,87].
6.	Australian Volunteers Program—Australia	Offers opportunities for Australians to volunteer overseas in countries across the Asia-Pacific region [85].

Note: source: [80–85].

6.5. Public–Private Partnership Collaboration

Implementing public–private partnerships (PPPs) in food waste reduction requires governments to establish a supportive framework. Governments can facilitate PPP formation by providing evidence-based support, funding, and enabling collaboration among stakeholders. Working with PPPs aligns with government policies and accelerates implementation while achieving a greater overall impact. The benefits for governments include cost savings, enhanced operational sustainability, improved food security, and significant carbon reductions, which support greenhouse gas policy objectives. Effective engagement with PPPs often spans multiple government departments and jurisdictions, necessitating cross-sector collaboration and stakeholder mapping to foster interest and shared responsibility for the success of the PPP [5].

Public–private partnerships (PPPs) play a pivotal role in addressing food waste, requiring signatories to agree on collective targets while evolving as membership expands (UNEP Food Waste Index Report 2024). Successful PPPs strategically include private organizations across the supply chain, with recruitment efforts being both deliberate and inclusive. Businesses are encouraged to engage with existing agreements or contribute to the establishment of new ones, viewing the cost as an investment. Studies show that every \$1 invested in food waste reduction can yield a \$14 return. Many companies are already innovating to transform food waste into business opportunities, such as the Kerry Group’s participation in the Australian Food Pact. Collaboration with governments is critical for the success of PPPs. Governments provide evidence, facilitate the formation of partnerships, and offer funding, benefiting from PPPs through cost savings, enhanced sustainability, and improved food security. These efforts align with policy objectives, while also reducing waste disposal costs. Given the complexity of food waste issues, PPPs often require cross-sector engagement, involving multiple government departments. Third parties, including NGOs and research institutions, play a vital role in enhancing the credibility of PPPs by providing neutral and independent advice. For instance, the Samen Tegen Voedselverspilling foundation in the Netherlands facilitates collaboration at various levels to reduce food waste effectively [5].

Public–private partnership (PPP) collaborations focus on promoting the minimization and control of food waste. The PPP approach not only reduces food waste in the food service sector, but also addresses waste across various other sectors, as highlighted by the UN Food Waste Index Report 2024 [5].

7. Discussion

The findings of this study highlight the dual benefits of food donation systems in the hospitality sector, presenting a sustainable solution to reduce food waste while simultaneously addressing the critical issue of food insecurity. The effectiveness of these systems lies in their capacity to efficiently redistribute surplus food to those in need, fostering both sustainability and social welfare. However, the study also identifies significant challenges in implementing these systems, underscoring the complexities involved. Key barriers include logistical issues, lack of awareness, insufficient collaboration, technological infrastructure limitations, and concerns about food quality and safety. Addressing these challenges is essential to enhancing the efficiency and impact of food donation systems within the hospitality sector [54].

To overcome these challenges, concerted efforts are needed from stakeholders, including the hospitality industry, non-profit organizations, and governmental bodies. Collaboration and knowledge sharing among these stakeholders are essential for developing robust logistical solutions and increasing awareness about the benefits and processes of food donation. Additionally, investing in technological infrastructure and adopting innovative digital platforms can greatly enhance the efficiency and effectiveness of food donation systems. Ensuring the quality and safety of donated food should remain a top priority, necessitating strict adherence to hygiene standards and regular monitoring. By addressing these challenges, the impact of food donation systems in the hospitality sector can be maximized, making a significant contribution to a sustainable future [51,56].

8. Conclusions

Optimizing the redistribution of surplus food in the hospitality sector through the implementation of food donation systems represents a paradigm shift toward greater responsibility for a sustainable future. This study highlights the profound impact of food donation systems in reducing food waste, alleviating food insecurity, enhancing stakeholder collaboration, and fostering sustainability within the hospitality industry. The research identifies and addresses key challenges, including logistical complexities, awareness gaps, collaboration inefficiencies, technological limitations, and food safety concerns. Overcoming these obstacles requires the adoption of advanced technologies, targeted awareness campaigns, stronger collaborative networks, and rigorous quality control measures.

The implementation of food donation systems in the hospitality sector was assessed through structured food donation models, technological interventions, and stakeholder collaborations. The study employed qualitative and quantitative approaches, including multiple regression analysis and hypothesis testing, to evaluate the system's effectiveness. The results indicate that food waste reduction was positively correlated with the effectiveness of food donation systems ($B = 0.164$, $p = 0.005$), confirming that structured donation programs significantly decrease surplus food waste. The promotion of sustainability showed a significant impact ($B = 0.204$, $p = 0.000$), emphasizing the role of donation programs in achieving the Sustainable Development Goals (SDG 2 and SDG 12). Partnership collaboration had the strongest influence ($B = 0.443$, $p = 0.000$), demonstrating that effective stakeholder engagement enhances food redistribution efficiency. The regression model confirmed that 55% of the variance in food donation effectiveness could be explained by the independent variables, proving the model's reliability in predicting successful implementation.

The outcome of the study was that the food donation system would streamline surplus food redistribution and enhance sustainability efforts through increased participation from hospitality practitioners. The obtained results validated these expectations, confirming that

hotels, restaurants, and catering services significantly contributed to reducing food waste while fostering corporate social responsibility. However, challenges in logistics, technological integration, and food safety compliance remained, indicating areas requiring further optimization. The successful implementation of food donation systems can significantly enhance the hospitality industry's reputation as a leader in sustainability. Beyond being a solution to food waste, these systems act as a transformative strategy that strengthens corporate social responsibility, improves food security, and supports environmental conservation efforts. By leveraging AI-based tracking, fostering stronger collaboration, and implementing policy-driven interventions, stakeholders can maximize the efficiency of food donation initiatives. Future efforts should focus on scaling food donation programs globally, integrating AI-driven logistics, and enhancing public–private partnerships to ensure seamless redistribution of surplus food. A data-driven approach, alongside real-time monitoring systems, can further improve the impact and sustainability of these programs.

9. Limitations and Further Research

The study was limited to the Alappuzha district of Kerala and focused exclusively on food services within the hospitality sector. The contribution of a graphical summary highlights the key findings. Future research could address long-term studies that track the effectiveness and impact of food donation systems over several years. Comparative studies across different geographic regions and cultural contexts could help identify best practices and region-specific challenges. Further research should also explore the development and integration of advanced technologies, such as AI-driven logistics and blockchain for traceability, in food donation systems. Examining their impact on efficiency, safety, and stakeholder engagement would provide valuable insights for optimizing these systems. Additionally, investigating the perceptions, attitudes, and motivations of various stakeholders—including hospitality businesses, non-profit organizations, and consumers—could offer a deeper understanding of the barriers and enablers of participation in food donation initiatives. These insights could inform targeted awareness and education campaigns to foster greater involvement and effectiveness.

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Data Availability Statement: The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Appendix A. Demographic Profile [64–74]

1. Age distribution of the respondents

- (a) 18 years to 28 years
- (b) 29 years to 39 years
- (c) 40 years to 50 years
- (d) 51 years to 61 years

- (e) 62 years and above
- 2. Gender of the respondents
 - (a) Male
 - (b) Female
 - (c) Other
- 3. Type of business
 - (a) Resorts
 - (b) Juice bars
 - (c) Bakeries
 - (d) Coffee shops
 - (e) Catering services
 - (f) Star hotels
 - (g) Casual hotels
 - (h) Cafes
 - (i) Restaurants
- 4. Designation of the respondents
 - (a) Owners
 - (b) Staff
 - (c) Chefs
 - (d) Managers

Instruction: the respondents were asked to indicate their level of agreement with the following statements using a 5-point Likert scale, where 1—strongly disagree, 2—disagree, 3—neutral, 4—agree, 5—strongly agree.

Appendix B. Factors Contributing to the Effectiveness of Food Donation System [28,64–74]

Section: Effectiveness of the food donation system

- (a) Our food donation process is efficient and well-organized.
- (b) The food donation system meets the intended social objectives.
- (c) Beneficiaries receive the donated food in good condition and on time.
- (d) We have seen measurable outcomes from implementing the food donation system.

A. Food waste reduction

- (a) Our organization actively tracks and monitors food waste.
- (b) Measures are taken regularly to minimize excess food generation.
- (c) The food donation system has reduced the amount of food wasted.
- (d) Staff members are trained in proper food storage and handling to prevent spoilage.

B. Promotion of sustainability

- (a) We consider food donation as part of our sustainability strategy.
- (b) Donating surplus food helps in achieving our environmental goals.
- (c) The initiative supports our commitment to responsible consumption.
- (d) Food donation contributes to our long-term sustainability goals.

C. Partnership collaboration

- (a) We collaborate with NGOs or food banks for food donations.
- (b) There is clear communication between our team and donation partners.
- (c) The success of our food donation system relies on strong partnerships.
- (d) Joint efforts with external organizations have improved donation effectiveness.

Appendix C. Implementation of the Food Donation System and Challenges [24,28,64–74]

Section: Implementation of the food donation system

- (a) Our organization has a structured process in place for food donations.
- (b) Clear guidelines and protocols are followed when donating surplus food.
- (c) The management actively supports the implementation of food donation initiatives.
- (d) The food donation system operates consistently, without major disruptions.
- (e) Staff members are aware of their roles and responsibilities in the donation process.
- (f) Our organization evaluates the performance of the food donation system regularly.
- (g) There is a designated team or individual responsible for overseeing food donations.
- (h) The system ensures that the food reaches beneficiaries in a timely and safe manner.

Logistical Challenges

- (a) Transportation of surplus food is a challenge for our organization.
- (b) We face delays or inefficiencies in distributing donated food.
- (c) Lack of proper storage facilities affects our ability to donate food.

Technology Infrastructure

- (a) We lack digital tools or software to manage food donations effectively.
- (b) There is no centralized system for tracking surplus food.
- (c) The absence of technological integration hampers our donation efforts.

Awareness

- (a) Staff members are not fully aware of food donation protocols.
- (b) There is a lack of training on the importance of food donation.
- (c) Many employees are unaware of partner organizations or food banks.

Collaboration

- (a) There is limited support from external partners or NGOs.
- (b) Lack of communication with stakeholders affects our donation process.
- (c) Difficulty in establishing reliable collaborations hinders implementation.

Quality and Safety

- (a) Concerns about food safety discourage donations.
- (b) We are unsure of legal liabilities related to donated food.
- (c) Maintaining food quality during transport is challenging.

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